Risk Management Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. Volatility, diversification, VaR and ES
2. Forward rates, duration and convexity
3. Immunization and asset-liability management
4. Currency and commodity risk
5. Greeks and delta hedging
6. Credit risk: default, loss and spreads
7. Counterparty risk and CVA
8. Real options and the Merton model
9. Operational risk and controls
10. Bank capital and liquidity rules
11. P vs Q, risk drivers and stress tests

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: Volatility, diversification, VaR and ES

VaR answers "how bad is a bad day?" Expected Shortfall answers "and if it gets worse than that, how bad on average?"

Start with an everyday idea

A weather forecast can tell you a storm threshold without saying how severe the worst storms are. VaR is a loss threshold; expected shortfall looks at the average loss beyond a tail threshold.

Words you need

Volatility describes variation. Diversification spreads exposure. VaR means value at risk for a stated time and confidence. ES means expected shortfall. Neither is a maximum possible loss.

Build the explanation

Volatility (σ): the standard deviation of returns. Bigger σ = wider swings.

Diversification: two assets that don't move together have a portfolio σ below the average of their σs. Correlation ρ controls how much.

VaR at 99%: the loss you will not exceed on 99 out of 100 days. For a normal distribution it is 2.326 × σ.

Expected Shortfall (ES): the average loss on the worst 1% of days. Always bigger than VaR, and it sees how fat the tail is. Regulators now prefer ES.

Formula and units

Portfolio variance: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2 \cdot w_1 \cdot w_2 \cdot \rho \cdot \sigma_1 \cdot \sigma_2$

Normal VaR: $VaR = z \times \sigma \times Value$ ($z = 1.645$ at 95%, 2.326 at 99%)

Normal ES: $ES = \sigma \times Value \times \varphi(z) / (1 - confidence)$

w = weights, ρ = correlation, φ = the normal bell-curve height.

Walk through the sketch

1. This bell curve is your daily profit and loss. Most days are near zero.
2. The red sliver on the left is the worst 1% of days.
3. VaR is where that sliver starts: a loss of $465k.
4. ES is the average of everything inside the sliver: $533k. Further out than VaR.
5. Diversification: mixing two assets gives 19.35% risk, less than the 24% average.

Worked example

A $10m portfolio has a daily σ of 2%. Find the 99% 1-day VaR and ES. Then: 60/40 in two assets with σ = 20% and 30%, ρ = 0.3. Portfolio σ?

1. Daily $σ = 0.02 × 10m = $200,000

2. VaR = 2.326 × 200,000 = $465,000

3. ES = 200,000 × φ(2.326)/0.01 = 200,000 × 2.665 = $533,000

4. $\sigma_p^2 = 0.36(0.04) + 0.16(0.09) + 2(0.6)(0.4)(0.3)(0.2)(0.3) = 0.03744$, $\sigma_p = 19.35\%$, below the 24% weighted average.

VaR $465k, ES $533k; portfolio σ 19.35%

Common mistakes

1. Calling VaR the worst possible loss. It is a threshold, not a maximum.

2. Scaling 1-day VaR to 10 days by 10 instead of √10.

3. Forgetting the 2·w1·w2 in the variance formula.

Try before checking

A one-day 95% VaR of $10 means, under the model…

1. Loss can never exceed $10

2. About 5% of days have losses beyond the threshold

3. The average daily loss is $10

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6123_INFO_01_Syllabus_Fall_2026.docx (RMN1); Risk lecture decks RMN1-2 are not in your folder yet

Lesson 2: Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

Start with an everyday idea

A fixed-rate bond is less attractive when new bonds pay more. Its price tends to fall. Duration estimates how strongly its price responds to a small rate change.

Words you need

Yield is a rate implied by price and payments. Modified duration measures price sensitivity. Convexity adjusts for curvature. A forward rate is a rate implied today for a future period.

Helpful earlier topics: Bonds and building a zero curve

Build the explanation

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always underestimates the price after a big move either way. Full repricing is the true answer; duration is the quick estimate.

Formula and units

Forward rate: $F(T1,T2) = (R2 \cdot T2 - R1 \cdot T1) / (T2 - T1)$

Duration: $D = \Sigma\ t \times [CF(t) \cdot e^{(-y \cdot t)}] / B$

Price change: $\Delta B \approx -B \times D \times \Delta y\ (+ \frac{1}{2} \times B \times C \times \Delta y^2)$

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Walk through the sketch

1. The price of the bond at each yield. It slopes down and bends (is "convex").
2. At today's yield of 12%, the price is $94.21.
3. Duration draws the tangent line at that point. Slope = $-B \times D$.
4. The tangent is below the curve on both sides. That gap is convexity.
5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

Worked example

3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.

2. Discount each at e^(-0.12t) and add: B = 94.213

3. Weight each time by its share of the price and add: D = 2.653 years

4. ΔB ≈ -94.213 × 2.653 × 0.001 = -0.250, so the price falls to about 93.963.

B = 94.21, D = 2.653, price drops about $0.25

Common mistakes

1. Using Δy = 0.1 instead of 0.001.

2. Forgetting the minus sign: rates up means price down.

3. Using duration for a large rate shock without the convexity term.

Try before checking

Modified duration is 5 and yield rises 0.01 (one percentage point). Approximate price change?

1. -5%

2. +5%

3. -0.05%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; Risk syllabus RMN2 (duration, convexity, term structure)

Lesson 3: Immunization and asset-liability management

A pension fund or bank owes money in the future. Immunization picks assets whose value moves the same way as those debts when rates change, so a rate move does not hurt.

Start with an everyday idea

If you owe money in the future, choose assets that respond to rates similarly to that obligation. This can reduce the effect of small rate changes on your ability to pay.

Words you need

A liability is an obligation. ALM means asset-liability management. Immunization matches value and rate sensitivity under assumptions; it does not remove every risk.

Helpful earlier topics: Bonds and building a zero curve, Forward rates, duration and convexity

Build the explanation

ALM (asset-liability management): manage assets and liabilities together, not separately. The risk is the gap between them.

Duration matching: make the PV of assets equal the PV of liabilities, and their durations equal. Then a small parallel rate shift changes both by about the same amount.

Convexity: also have asset convexity ≥ liability convexity, so bigger moves help rather than hurt.

Rebalance: durations drift as time passes and rates move, so immunization must be redone.

Key-rate duration: the curve does not always shift in parallel. Key-rate durations measure exposure to each part of the curve (2y, 5y, 10y…) separately.

Hedge accounting (RMN8): rules that let a company show a hedge and the item it hedges together in its accounts, so the P&L does not jump around.

Formula and units

Match PV: $PV(\text{assets}) = PV(\text{liabilities})$

Match duration: $w_1 \cdot D_1 + w_2 \cdot D_2 = D_L$, $w_1 + w_2 = 1$

Surplus change: $\Delta S \approx -(A \cdot D_A - L \cdot D_L) \times \Delta y$

w = share of money in each bond, D = duration, A and L = PV of assets and liabilities.

Walk through the sketch

1. A timeline. You owe $100m in present-value terms at year 5.

2. You can only buy bonds that pay at year 2 and year 10.

3. Think of a seesaw balanced at year 5: 62.5% at year 2 and 37.5% at year 10 balance it.

4. Now if rates move a little, assets and the liability move by about the same amount. You are immunized.

Worked example

You owe $100m (PV) in 5 years. You can buy 2-year and 10-year zero-coupon bonds. How much of each?

1. Zero-coupon bond duration = its maturity: D1 = 2, D2 = 10.

2. Solve w1 × 2 + w2 × 10 = 5 with w1 + w2 = 1.

3. w2 × 8 = 3 → w2 = 0.375, w1 = 0.625

4. Buy $62.5m of 2-year zeros and $37.5m of 10-year zeros (in PV). This "barbell" also has more convexity than the liability.

62.5% in 2-year, 37.5% in 10-year

Common mistakes

1. Matching maturities instead of durations.

2. Forgetting to also match PV.

3. Assuming the hedge lasts forever; durations drift and must be rebalanced.

Try before checking

Matching duration is intended mainly to reduce…

1. Small interest-rate-shock risk

2. Every possible loss

3. All default risk

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN2 and RMN8 (duration, immunization, ALM, hedge accounting); FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx

Lesson 4: Currency and commodity risk

Exchange-rate forwards follow interest rate parity: the currency with the higher interest rate trades at a discount forward. Commodity forwards add storage costs and a convenience yield.

Start with an everyday idea

You will pay euros next month but earn dollars. Even if the euro bill stays the same, its dollar cost can change before you pay.

Words you need

FX means foreign exchange. Exposure is the amount affected by a risk. Commodity carry includes financing and storage, offset by benefits of holding the physical good.

Helpful earlier topics: Forward price by no-arbitrage

Build the explanation

FX quote: EUR/USD = 1.10 means 1 euro costs 1.10 dollars.

Covered interest rate parity: you cannot earn free money by borrowing in one currency, converting, investing in the other, and locking in the rate back. So F = S·e(r-rf)T.

FX exposure: a US firm with euro revenue loses if the euro falls. Hedge by selling euros forward.

Commodities: holding oil costs storage (u). But having it on hand also has a benefit, the convenience yield (y). F = S·e(r+u-y)T.

Contango: F > S (storage dominates). Backwardation: F Equity risk (also RMN3): value a stock with dividends or factors (CAPM, multi-factor).

Formula and units

FX forward: F = S × e^((r - r_f)·T)

Commodity forward: F = S × e^((r + u - y)·T)

S = spot price (in domestic currency per unit), r = domestic rate, r_f = foreign rate, u = storage cost rate, y = convenience yield.

Walk through the sketch

1. Path 1: keep $1.10 in dollars at 5% for a year. You end with 1.10 × e^0.05.

2. Path 2: change it to €1, earn 3% in euros, and convert back at the forward rate F.

3. Both paths must give the same dollars, or there is free money. So F = 1.10 × e^0.02 = 1.1222.

4. Commodities: storage pushes forward prices up (contango); a convenience yield pulls them down (backwardation).

Worked example

EUR/USD spot is 1.10. USD rate 5%, EUR rate 3% (continuous). What is the 1-year forward? A US exporter will receive €1m in a year. What can it lock in?

1. F = 1.10 × e^((0.05 - 0.03) × 1) = 1.10 × e^0.02

2. e^0.02 = 1.0202 → F = 1.1222 USD per EUR

3. The euro is at a premium because its interest rate is lower.

4. Sell €1m forward: locks in $1,122,200 no matter where the euro goes.

F = 1.1222; locks in $1.1222m

Common mistakes

1. Flipping r and r_f (check: the lower-rate currency is at a forward premium).

2. Mixing up which currency the quote is "per".

3. Leaving out convenience yield when the futures curve is downward sloping.

Try before checking

You owe €100. EUR/USD rises from 1.10 to 1.20. Dollar cost…

1. Falls $10

2. Rises $10

3. Does not change

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN3 (FX exposures and parity, equity, commodity carry and basis); FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx

Lesson 5: Greeks and delta hedging

A trader who sells options removes most of the risk by holding delta shares of the stock, then adjusting as delta changes. Gamma says how often you need to adjust.

Start with an everyday idea

If an option moves about half as much as its stock, half a share can offset that small price movement in a suitable opposite position. But the relationship changes, so hedges need updating.

Words you need

Delta is first-order price sensitivity. Gamma measures how delta changes. Rebalancing means adjusting the hedge. Delta-neutral means near-zero local sensitivity, not zero risk.

Helpful earlier topics: Black-Scholes and the Greeks, Hedging with futures

Build the explanation

Delta-neutral: total delta of the book = 0, so small stock moves do not change its value.

Sold calls have negative delta, so you buy shares to hedge. Sold puts have positive delta, so you sell shares.

Gamma: delta changes as the stock moves. New delta ≈ old delta + $\Gamma \times \Delta S$. High gamma = rebalance often.

A short-option, delta-hedged book has negative gamma: it loses on big moves, and earns theta (time decay) on quiet days.

Gamma/vega hedging: stock has zero gamma and vega, so you need other options to hedge those.

Hedging error: you only rebalance at discrete times and volatility is not what you assumed, so the hedge is never perfect.

Formula and units

Shares to hold = -(option position) × Δ

New $\Delta \approx \Delta + \Gamma \times \Delta S$

P&L of hedged short option $\approx \theta \cdot \Delta t - \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$ (per option, sign for a short)

Δ = delta, Γ = gamma, θ = theta (here per unit time, positive for a short option).

Walk through the sketch

1. Delta of a call (K = 40) against the stock price. It goes from 0 to 1.

2. At S = 42 delta is about 0.78. Short 1 call → buy 0.78 shares.

3. The stock rises to 46. Delta has climbed (that climb is gamma). You must buy more shares.

4. The steepest part of the curve is near the strike: that is where gamma is highest and you rebalance most.

Worked example

You sold 1,000 calls with delta 0.60 and gamma 0.04. How many shares to buy? The stock then rises $1.25. What do you do?

1. Option delta = -1,000 × 0.60 = -600 → buy 600 shares

2. New delta ≈ 0.60 + 0.04 × 1.25 = 0.65

3. You now need 1,000 × 0.65 = 650 shares

4. Buy 50 more. (Notice: you buy after a rise and sell after a fall. That is how negative gamma costs money.)

Buy 600 shares, then 50 more

Common mistakes

1. Selling shares to hedge a short call (it is buy).

2. Thinking delta-neutral means no risk; gamma and vega remain.

3. Forgetting the contract multiplier (1 contract often = 100 shares).

Try before checking

A short call has delta -0.6 per share unit. A local hedge is…

1. Buy 0.6 share units

2. Sell 0.6 share units

3. Ignore the stock

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Hull 8e, Chapter 18 (The Greek letters); Risk syllabus RMN4 (Greeks, dynamic hedging)

Lesson 6: Credit risk: default, loss and spreads

Expected credit loss is chance of default × share you lose × amount at stake. A credit spread is the extra yield lenders demand to cover that.

Start with an everyday idea

A friend may fail to repay a loan. How often that happens, how much you would lose, and how much is owed are three separate questions.

Words you need

PD is probability of default. LGD is the fraction lost if default happens. EAD is exposure at default. Recovery is the fraction you get back.

Helpful earlier topics: Money today vs money later, Volatility, diversification, VaR and ES

Build the explanation

PD (probability of default), LGD (loss given default = 1 - recovery rate), EAD (exposure at default).

Hazard rate λ: the default "speed". Survival to time t is e-λt; PD by t is 1 - e-λt.

Credit spread ≈ λ × LGD. A CDS (credit default swap) is insurance on a bond, and its spread works out to about the same number.

P vs Q: real-world (P) default probabilities come from history. Risk-neutral (Q) ones come from prices and can differ because prices reflect risk premia and other market effects.

Expected loss is a cost you price in. Unexpected loss is what capital must absorb.

Formula and units

Survival: S(t) = e^(-λt) PD(t) = 1 - e^(-λt)
Expected loss: EL = PD × LGD × EAD
Spread: s ≈ λ × LGD → λ ≈ s / (1 - R)

λ = hazard rate per year, R = recovery rate.

Walk through the sketch

1. The chance a borrower is still alive (not defaulted) over time.
2. With λ = 2% a year, survival slides slowly down: e^(-0.02t).
3. At 5 years, 90.48% survive. So 9.52% have defaulted.
4. Expected loss stacks three things: $1m at stake × 60% lost × 1.98% chance = $11.9k.

5. The lender wants about λ × LGD = 1.2% extra yield to cover this.

Worked example

A borrower has λ = 2% a year. LGD = 60%. Exposure $1m. Find 5-year survival, 5-year PD, 1-year expected loss and the spread.

1. Survival 5y = e^(-0.02 × 5) = e^(-0.1) = 90.48%

2. PD 5y = 1 - 0.9048 = 9.52%

3. 1-year PD ≈ 1 - e^(-0.02) = 1.98%, EL ≈ 0.0198 × 0.6 × 1,000,000 = $11,900

4. Spread ≈ 0.02 × 0.6 = 1.2% = 120 basis points

Survival 90.48%, PD 9.52%, EL ≈ $11.9k, spread ≈ 120 bp

Common mistakes

1. Using recovery rate where LGD is needed (or the reverse).

2. Adding PDs across years instead of using survival.

3. Mixing P and Q probabilities in the same calculation.

Try before checking

PD 2%, LGD 50%, EAD $1,000. Expected loss?

1. $10

2. $20

3. $500

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf; Risk syllabus RMN5-6 (PD, LGD, CDS, CVA); Hull 8e, Chapter 23 (Credit risk)

Lesson 7: Counterparty risk and CVA

If the other side of a derivative defaults while it owes you money, you lose. CVA is the price of that risk, built from exposure, default probability and loss given default.

Start with an everyday idea

A contract can be profitable on paper yet fail to pay if the other party defaults. CVA reduces its value to reflect that possibility.

Words you need

Counterparty is the other party to a contract. Exposure is what you stand to lose if they default. CVA means credit valuation adjustment. Collateral can reduce exposure.

Helpful earlier topics: Credit risk: default, loss and spreads

Build the explanation

Exposure: only positive value matters. If the trade is worth +5 to you and they default, you lose up to 5. If it is worth -5, you still owe them.

Expected exposure EE(t): the average of max(V, 0) across simulated futures. For a swap it rises then falls (a hump), because fewer payments remain later.

PFE (potential future exposure): a high percentile (e.g. 95%) of exposure. Used for limits.

CVA (credit valuation adjustment): expected loss from counterparty default, subtracted from the trade value.

Reducing it: netting, collateral (margin), and central clearing. xVA adds related adjustments (DVA, FVA, KVA).

Migration: a counterparty can be downgraded before defaulting, which raises CVA too.

Formula and units

Exposure: $E(t) = \max(V(t), 0)$

$\text{CVA} \approx \text{LGD} \times \Sigma\ \text{EE}(t_i) \times \text{PD}(t_{i-1}, t_i) \times \text{DF}(t_i)$

PD(t_{i-1}, t_i) = probability of default in that period, DF = discount factor.

Walk through the sketch

1. Possible future values of a swap. Some paths end up in your favour (above 0), some against you.

2. Only the part above zero is at risk. Averaging that gives the expected exposure: a hump over time.

3. Multiply each year's exposure by the chance of default in that year.

4. Add them up and multiply by LGD: CVA ≈ $72k.

Worked example

A 3-year swap has expected exposure $2m, $3m, $1m in years 1, 2, 3. The counterparty has a 2% chance of defaulting in each year; LGD 60%. Ignore discounting. CVA?

1. Year 1: 2 × 0.02 = 0.04

2. Year 2: 3 × 0.02 = 0.06; Year 3: 1 × 0.02 = 0.02

3. Sum = 0.12, times LGD 0.6 = 0.072

4. CVA ≈ $72,000. The trade is worth $72k less than if the counterparty were risk-free.

CVA ≈ $72,000

Common mistakes

1. Using V instead of max(V, 0).

2. Using cumulative PD for every year instead of the PD within each period.

3. Forgetting that collateral and netting cut exposure.

Try before checking

What does CVA account for?

1. The counterparty failing to pay

2. Only stock-price volatility

3. A guaranteed profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN5-6 (CVA, exposure profiles, xVA); Hull 8e, Chapter 24 (Credit risk)

Lesson 8: Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

Start with an everyday idea

A business can wait before building a factory, just as an option holder can wait before exercising. Flexibility can have value when the future is uncertain.

Words you need

A real option is flexibility over a business decision. Merton models a company’s equity like a call on its assets, with debt due at a future date.

Helpful earlier topics: The four derivatives and their payoffs, Credit risk: default, loss and spreads, Black-Scholes and the Greeks

Build the explanation

Real option: the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.

Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.

If the option value is bigger than investing now (NPV today), wait.

Energy example (VL14): a power plant is a call on the spread between power and fuel prices: run it only when profitable.

Merton model: equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. Distance to default counts how many σ the firm is from that line.

Formula and units

Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)

Option value = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)

Merton: Equity = V·N(d1) - D·e^(-rT)·N(d2)

DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T), PD = N(-DD)

V = value of project or firm assets, I = investment cost, D = face value of debt.

Walk through the sketch

1. Today the project is worth $105 and costs $100. NPV = $5.

2. Next year it is worth $140 (good) or $80 (bad).

3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.

4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.

5. Same idea for a firm: shareholders get max(Assets - Debt, 0). Equity is a call on the assets.

Worked example

A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 - 100 = 5

2. p = (105 × 1.05 - 80) / (140 - 80) = 30.25/60 = 0.504

3. Wait: next year invest only if up: payoff 40 or 0

4. Option value = 0.504 × 40 / 1.05 = 19.21. Bigger than 5, so wait.

5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = 1.01, PD = N(-1.01) ≈ 15.7%

Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common mistakes

1. Using the real-world probability in the roll-back.

2. Comparing the option with zero instead of with investing now.

3. In Merton, using the stock's volatility as the asset volatility.

Try before checking

A factory project can wait one year for more information. That flexibility…

1. May have value

2. Must be worthless

3. Guarantees profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf; FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; Risk syllabus RMN6 (Merton, distance to default)

Lesson 9: Operational risk and controls

Operational risk is loss from failed people, processes, systems or outside events: a fat-finger trade, a hack, a payment sent twice, fraud.

Start with an everyday idea

A correct trading idea can still lose money if someone enters an extra zero or a system fails. Operational risk is about failures in people, processes, systems and external events.

Words you need

A control prevents or detects mistakes. Reconciliation compares independent records. A KRI is a key risk indicator: a warning signal, not proof that a loss will happen.

Build the explanation

Basel definition: loss from inadequate or failed internal processes, people and systems, or from external events. Includes legal risk.

Frequency × severity: many small losses (errors) and rare huge ones (fraud, rogue trading). The rare ones dominate the tail.

Controls: preventive (stop it happening: approvals, access limits) and detective (catch it: reconciliations, alerts).

KRIs (key risk indicators): early-warning numbers, e.g. failed trades per day, staff turnover, overdue reconciliations.

Three lines of defense: 1) the business owns the risk, 2) risk & compliance oversee, 3) internal audit checks independently.

Payments and settlement risk: paying before you get paid (Herstatt risk). Fixed by delivery-versus-payment and CLS for FX.

Technology/cyber risk and model risk sit here too.

Formula and units

Expected annual loss ≈ expected frequency × expected severity

Residual risk = inherent risk - effect of controls

Op-risk capital looks at the tail of the loss distribution, not just the average.

Walk through the sketch

1. A bow-tie: the risk event sits in the middle (e.g. a payment sent twice).
2. On the left are the causes: people, process, systems, external events.
3. Preventive controls sit between causes and the event. They try to stop it.
4. On the right are the consequences. Detective controls and recovery plans limit the damage.

Worked example

A desk has about 12 settlement errors a year, each costing $50,000 on average. A new reconciliation control would catch 75% of them and costs $200,000 a year. Is it worth it (just on expected loss)?

1. Expected loss now: 12 × 50,000 = $600,000 a year
2. Losses avoided: 75% × 600,000 = $450,000
3. Net benefit: 450,000 - 200,000 = $250,000 a year
4. Yes, and it also cuts tail risk (fewer chances of one giant error).

Worth it: saves about $250,000 a year

Common mistakes

1. Looking only at average losses and ignoring rare severe events.
2. Confusing a control (an action) with a KRI (a measurement).
3. Treating operational risk as only IT risk.

Try before checking

Which is an operational-risk control?

1. Two-person approval for a large payment
2. Predicting stock prices
3. Buying only high-return assets

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN7 (operational risk, payments, controls, KRIs, technology risk)

Lesson 10: Bank capital and liquidity rules

Banks need capital to absorb losses and liquid assets to meet payments. The ratios below are simplified Basel-style classroom examples, not a bank-specific compliance assessment.

Start with an everyday idea

A bank needs both a cushion to absorb losses and cash-like assets to meet near-term payments. Being solvent and being able to pay today are related but different.

Words you need

Capital absorbs losses. Liquidity means meeting payments when due. RWA means risk-weighted assets. Course ratio examples are simplified; applicable rules depend on bank and jurisdiction.

Helpful earlier topics: Volatility, diversification, VaR and ES, Credit risk: default, loss and spreads

Build the explanation

Capital: equity that can absorb losses. CET1 (common equity tier 1) is the best quality.

Risk-weighted assets (RWA): each asset is weighted by riskiness (cash ~0%, mortgages lower, corporate loans ~100%). Includes market and operational risk too.

Illustrative baseline ratios: CET1 ≥ 4.5% of RWA, Tier 1 ≥ 6%, total capital ≥ 8%, plus a 2.5% conservation buffer (and extra for big banks).

Leverage ratio: Tier 1 / total exposure (not risk weighted) ≥ 3%. A backstop if risk weights are wrong.

LCR (liquidity coverage ratio): high-quality liquid assets / net cash outflows over 30 days of stress ≥ 100%.

NSFR (net stable funding ratio): stable funding / required stable funding ≥ 100% over one year.

Formula and units

CET1 ratio = CET1 / RWA (≥ 4.5%, ≥ 7% with buffer)
Leverage = Tier 1 / Total exposure (≥ 3%)
LCR = HQLA / 30-day net outflows (≥ 100%)

HQLA = high-quality liquid assets (cash, central bank reserves, government bonds).

Walk through the sketch

1. A bank balance sheet: assets on the left, how they are funded on the right.

2. Most funding is deposits and debt. A thin slice is equity: that is capital.

3. Weight the assets by risk to get RWA. Capital must be at least 7% of that (with the buffer).

4. Liquidity is a separate test: enough liquid assets to cover 30 days of stressed outflows.

Worked example

A bank has CET1 $60bn, RWA $800bn, total exposure $1,500bn, HQLA $120bn and stressed 30-day outflows of $100bn. Assume Tier 1 = CET1. Does it pass?

1. CET1 ratio = 60 / 800 = 7.5% ≥ 7% ✓

2. Leverage = 60 / 1,500 = 4.0% ≥ 3% ✓

3. LCR = 120 / 100 = 120% ≥ 100% ✓

4. It passes all three, but only 0.5% above the CET1 buffer: a $4bn loss would reach that threshold if RWA stayed unchanged; a larger loss would breach it.

7.5% CET1, 4% leverage, 120% LCR: passes

Common mistakes

1. Dividing capital by total assets for the CET1 ratio (use RWA).

2. Thinking capital is cash. Capital is how the bank is funded, not what it holds.

3. Mixing up liquidity (cash to pay today) and solvency (enough capital overall).

Try before checking

A bank has valuable long-term assets but cannot meet withdrawals today. The immediate problem is…

1. Liquidity

2. A high stock return

3. Too much profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN9 (bank balance sheet, capital, leverage, liquidity); BIS Basel Framework (see "Watch online")

Lesson 11: P vs Q, risk drivers and stress tests

P is the real world (what actually tends to happen). Q is the risk-neutral world used for pricing. Risk managers measure losses in P, price in Q, and stress-test both.

Start with an everyday idea

There are two different questions: “What might really happen?” and “What price is consistent with today’s traded assets?” They need different probability tools.

Words you need

P labels real-world probabilities used for forecasting/risk. Q labels risk-neutral pricing probabilities. A stress test asks what happens under a chosen severe scenario, without needing to call it likely.

Helpful earlier topics: Volatility, diversification, VaR and ES, Binomial trees for options

Build the explanation

Q (risk-neutral): probabilities that make every asset earn the risk-free rate. Used for pricing (binomial trees, Black-Scholes, CDS). Not a forecast.

P (physical/real world): probabilities from history and judgement. Used for VaR, expected losses, capital.

Pricing weights differ from real-world probabilities. In many credit examples, implied Q default rates exceed historical rates because spreads include risk premia, but this is not a universal ordering of all outcomes.

Risk drivers: the few market factors that move a portfolio (rates, spreads, equity index, FX, vol). Map each position to them.

Aggregation: combine risks across desks using correlations; diversification lowers the total, but correlations jump in a crisis.

Ex-ante P&L / stress tests: apply a scenario (e.g. rates +200bp, equities -30%, spreads +150bp) and reprice everything to see the loss before it happens.

Formula and units

Q default intensity from spread: $\lambda Q \approx s / LGD$

Stress loss $\approx \Sigma$ (sensitivity to driver $\times$ shock) (or full revaluation)

Bond: $\Delta V \approx -V \times D \times \Delta y$

Use P for "how likely / how much could we lose", Q for "what is it worth".

Walk through the sketch

1. P: the real-world spread of outcomes for a risky asset. Its average is above the risk-free rate.

2. Q: the same outcomes reweighted so the average is the risk-free rate. More weight on bad outcomes.

3. Use P for risk (VaR, expected loss). Use Q for prices.

4. Stress test: shock each driver and add the hits. Bonds -$6m, equities -$9m, total -$15m.

Worked example

A bond portfolio worth $50m (duration 6) and an equity portfolio worth $30m. Stress: rates +2%, equities -30%. Loss? Also: a spread of 150bp with 60% LGD implies what Q default rate, if history says 1%?

1. Bonds: -50 × 6 × 0.02 = -$6m

2. Equities: -30% × 30 = -$9m

3. Total stress loss ≈ $15m (19% of $80m)

4. λQ ≈ 0.015 / 0.6 = 2.5% a year vs 1% in P. The gap is the credit risk premium.

Stress loss ≈ $15m; Q default rate 2.5% vs P 1%

Common mistakes

1. Using Q probabilities to forecast real default rates.

2. Adding stress losses that cannot happen together, or ignoring ones that do.

3. Trusting normal-times correlations in a stress scenario.

Try before checking

Use risk-neutral probabilities mainly to…

1. Predict the actual frequency of every event

2. Price claims consistently under a model

3. Guarantee no losses

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMNA (P vs Q, drivers, aggregation, ex-ante P&L, stress); Valuation notes: instructor-linked explanation of risk neutrality

Lesson-check answers

1. Volatility, diversification, VaR and ES

Answer: About 5% of days have losses beyond the threshold. 95% of modeled outcomes are at or below the loss threshold. The remaining tail can contain much larger losses.

2. Forward rates, duration and convexity

Answer: -5%. Approximate proportional change = -5 × 0.01 = -0.05, or -5%. This is an estimate, not exact repricing.

3. Immunization and asset-liability management

Answer: Small interest-rate-shock risk. It targets rate sensitivity. Large or non-parallel curve moves, defaults and changing cash flows can still hurt.

4. Currency and commodity risk

Answer: Rises $10. Old cost = 100 × 1.10 = $110. New cost = 100 × 1.20 = $120.

5. Greeks and delta hedging

Answer: Buy 0.6 share units. The stock contributes +0.6 delta, offsetting -0.6. Contract multipliers determine the actual share count.

6. Credit risk: default, loss and spreads

Answer: $10. 0.02 × 0.50 × 1,000 = $10. This average is not the loss in every outcome.

7. Counterparty risk and CVA

Answer: The counterparty failing to pay. CVA reflects expected loss from counterparty default, combining exposure, default probabilities and loss severity.

8. Real options and the Merton model

Answer: May have value. Waiting can avoid a bad investment, though delay can also lose revenue. The value depends on assumptions.

9. Operational risk and controls

Answer: Two-person approval for a large payment. Independent approval can catch errors or unauthorized payments before money moves.

10. Bank capital and liquidity rules

Answer: Liquidity. It may have enough asset value overall but insufficient accessible cash. Capital and liquidity solve different problems.

11. P vs Q, risk drivers and stress tests

Answer: Price claims consistently under a model. Q is a pricing device. Real-world frequencies and risk forecasts use P; neither eliminates uncertainty.

Prepare a risk report, not a guessed final exam

The supplied FRE 6123 syllabus assigns 40% to team assignments and 30% each to two individual report/presentation assignments. The RMN lecture decks were absent from the supplied pack; these notes are foundations, not substitutes for those decks.

A report rehearsal

1. Choose one clearly bounded risk exposure and decision-maker.

2. State the data source, period, units and key assumptions.

3. Explain one suitable risk measure and compute a small transparent example.

4. Stress an assumption, discuss limitations and propose a response.

5. Prepare a short presentation where every chart answers a decision question. Follow the instructor’s actual assignment brief.